

Gauss–Legendre quadrature

1 Rule

For $f \in C[-1, 1]$ the n -point Gauss–Legendre rule is

$$\int_{-1}^1 f(x) \, dx \approx \sum_{i=1}^n w_i f(x_i), \quad (1)$$

where $x_1 < \cdots < x_n$ are the zeros of the Legendre polynomial P_n and

$$w_i = \frac{2}{(1 - x_i^2) [P_n'(x_i)]^2}. \quad (2)$$

Theorem 1. *Rule (1) is exact for all polynomials of degree at most $2n - 1$, and for $f \in C^{2n}[-1, 1]$ there is $\xi \in (-1, 1)$ with*

$$\int_{-1}^1 f(x) \, dx - \sum_{i=1}^n w_i f(x_i) = \frac{2^{2n+1} (n!)^4}{(2n+1) [(2n)!]^3} f^{(2n)}(\xi).$$

2 Computation

The nodes are the eigenvalues of the symmetric tridiagonal Jacobi matrix J_n with $(J_n)_{i,i+1} = (J_n)_{i+1,i} = \beta_i = i/\sqrt{4i^2 - 1}$, and $w_i = 2v_{1i}^2$ for the normalized eigenvectors v_i of J_n [1].

References

- [1] G. H. Golub, J. H. Welsch, Calculation of Gauss quadrature rules, Math. Comp. 23 (1969) 221–230.